Firozabad Survey Progress Report

Week: 18/09/2026 to 25/09/2026

1. **Survey Overview**

Survey Component	Total	Completed this week (number and %)	Planned Coming Week	Areas/Locations Covered
DGPS Survey of Drains [As per OSM data subject to change]	666.869 kms	24.2 kms	Mapping of Secondary drains	Suhag Nagar Road, Lalao Road, New Bypass Road, Nala Road, Azad Nagar
Drone Survey Area	35.10 sq.kms	Got the permission from the Police Department and covered 14 sq.kms till 24/09/2026 (39.8%)	-	**Blocks Covered:** FRD_03 and FRD_01 **Ground Control / Base Points**: FRD.CP-15, FRD.ST-05, and FRD.CP-29

2. **Survey Progress till 17th September**
   - Survey started at Firozabad on 28th August and was carried out till 9th September. In the due course of time the following activities were completed.
   - A total of **89** GCP's for drone survey have been marked.
   - Base points for DGPS Survey - **5**
   - DGPS survey of drains- **2.705 kms done** (Area covered so far: DGPS- Fatehabad Road, Firozabad)
   - The Firozabad survey team discontinued field activities due to unsuitable working conditions, the works were started again from 17th September.

3. DETAILS OF SURVEY TEAM & DGPS/ROVER EQUIPMENT-FIROZABAD

Date	DGPS Base	Rover	Drone	Members
18-09-2026	1	1	-	2
19-09-2026	1	1	-	2
20-09-2026	1	1	-	2
21-09-2026	1	1	-	2
22-09-2026	1	1	-	2
23-09-2026	1	1	-	2
24-09-2026	1	1	1	3
25-09-2026	1	1	1	3

4. DGPS Survey of Drains

DGPS-based cross-section survey of the Nagar Nigam drain network commenced on 08 September 2026 and has continued through 25 September 2026, carried out by the field survey teams together with the Firozabad base crew. The survey includes capturing critical cross sectional and hydraulic features such as **drain top and bottom edges, bed left and right, bank top left and right, manhole rim levels, culverts, water level and nala top and bed points**, referenced to temporary benchmarks (TBM) established from the project base station.

Date	Stretch covered (km)
18-09-26	1.8
19-09-26	5
22-09-26	8.5
23-09-26	8.9
24-09-26	7
25-09-26	9.5
Total	**40.7**

5. Lidar Survey

Drone permission was granted to Firozabad on 19th September by Police Department. On 24th September 2026, drone survey operations were carried out across Blocks FRD_03 and FRD_01, during which a total of six flights were conducted. Approximately 14 sq.km of area was covered, with base points established at FRD.CP-15, FRD.ST-05, and FRD.CP-29. The survey commenced at 9:50 AM and concluded at 5:30 PM. It may be noted that partial cloud cover was observed during the survey period. On 25th September due to windy conditions and rains the drone survey was not done.

6. Reconnaissance Survey

A reconnaissance survey was conducted in Firozabad from **17th to 19th September**, covering major stormwater drains, nalas, outfalls, STP's, pumping stations and key water bodies across the city. The **modelling team, along with the project SPOC, Mr. Raghavendra Singh**, carried out field inspections to understand the existing drainage conditions, flow paths, outfall connectivity, and potential waterlogging locations. Key pumping stations and STP's were observed along with the Firozabad Nagar Nigam official. Geotagged photographs and field observations were collected to support the subsequent hydrological and hydraulic modelling and assessment.

Survey Programme

Day	Date	Area / Corridor	Major Activities
Day 1	17-Sep-26	New Bypass Road	Reconnaissance of the Pampa Nehar Canal, covering approximately 8.81 km within the study boundary, including upstream-to-downstream inspection and assessment of canal condition and drainage usage.
Day 2	18-Sep-26	Suhag Nagar Road, Near RG Cassette Company Road, New Bypass Road Azad Nagar	Field inspection of two primary drainage channels, covering their respective stretches from upstream to downstream, with assessment of drainage condition, sedimentation, waste accumulation and crossing structures.
Day 3	19-Sep-26	Om Nagar Colony, Chandravan Road, Sanjay Singh Colony, Chandravan Road, Indon, Edhon	Inspection of secondary drains and their connectivity with primary drains, followed by field verification of the downstream outfall, pumping stations and STP's leading towards the Yamuna River.

S. No	Location	Co-ordinates	Field Observation
1	New Bypass Road (Start of Canal as per Boundary)	27°11'15.27"N 78°23'8.37"E	Dense vegetation and weed growth are obstructing the open drainage channel. Stagnant/slow-moving and visibly polluted water is present within the drain.
2	Near Shri Vaishnavi Traders, New Bypass Road,	27°10'37.05"N 78°23'49.65"E	The drain is heavily covered with algal growth and stagnant water, indicating poor flow conditions and water quality deterioration.
3	New Bypass Road	27° 9'15.76"N 78°25'14.09"E	Wastewater accumulated in the adjacent low-lying area is being discharged into the main drainage channel.
4	Near Mankameshwari Mandir (End of Canal as per boundary)	27° 7'33.05"N 78°25'20.50"E	The drainage channel is open and unlined, with an irregular cross-section and naturally formed side slopes. .
5	Near Rachana Vegetable Shop, Suhag Nagar Road	27° 9'24.05"N, 78° 22'32.91"E	The cross-drainage structure has an exposed and partially deteriorated deck, with no proper parapet or edge protection along the crossing.
6	New Bypass Road Azad Nagar	27°09' 00.72" N, 78° 25' 25.39" E	Drain water is being abstracted through a temporary pumping arrangement and reused for construction and cleaning activities. The drain water appears turbid and contaminated, with visible suspended and floating impurities.
7	Near Railway Stations Culvert, Suhag Nagar Road.	27° 9'5.30"N, 78°22'37.29"E	The drainage channel near the railway station carries high-flow wastewater with visible foaming, indicating significant wastewater inflow and poor water quality.
8	Near RG Cassette Company Road	27° 8'1.25"N, 78°23'45.71"E	The box culvert inlet is heavily obstructed by accumulated solid waste, significantly restricting the effective flow opening.

7. Instrument Calibration

The DGPS equipment deployed for the survey has been calibrated and is within the validity period. The calibration certificates have been submitted as supporting documentation for the survey equipment.

8. Report Status

Report	Status	Comments
Inception Report	Report submitted to Project Manager	Sent on 16th September 2026

9. Static Points



Figure 1Near Sofipur Tiraha, Fathhabad road, FRD_ST-01



Figure 2 Near Suhar Nagar Road, FRD_ST-02





Figure 3 Near Raipura, FRD_ST-03



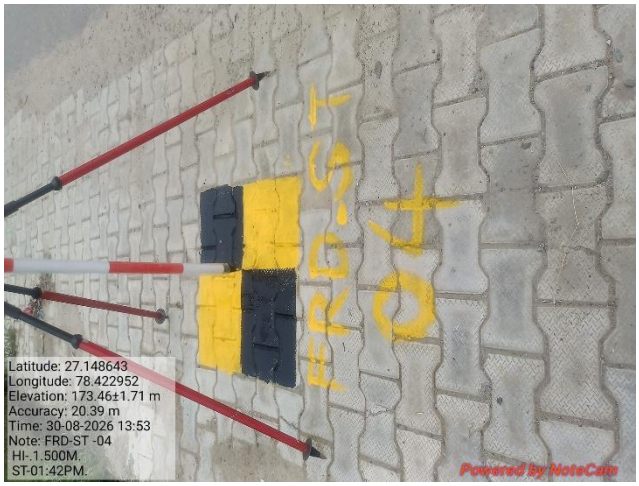


Figure 4 Near Selai Road (Conjunction of Selai Road and New Bypass Road), FRD_ST-04





Figure 5 Near Ramleela Ground FRD_ST-05

10. Site Survey Photographs of Second Week (Till 23rd September)










UTM: 44R
242775mE 3004224mN
Elevation: 169.08±2.5 m
Accuracy: 3.79 m
Time: 21-09-2026 10:27
Note: FRD.CP-75
Powered by NoteCam


UTM: 44R
246954mE 3003447mN
Elevation: 174.78±2.5 m
Accuracy: 3.79 m
Time: 21-09-2026 08:39
Note: TBM.1
Powered by NoteCam


UTM: 44R
244028mE 3006555mN
Elevation: 175.84±2.5 m
Accuracy: 3.79 m
Time: 22-09-2026 15:41
Powered by NoteCam


UTM: 44R
240010mE 3005246mN
Elevation: 154.78±2.5 m
Accuracy: 3.79 m
Time: 22-09-2026 09:59
Powered by NoteCam


UTM: 44R
246951mE 3003451mN
Elevation: 176.39±2.5 m
Accuracy: 3.79 m
Time: 23-09-2026 17:43
Note: TBM.1
Powered by NoteCam


UTM: 44R
241762mE 3008640mN
Elevation: 182.99±2.5 m
Accuracy: 3.79 m
Time: 23-09-2026 10:11
Powered by NoteCam


UTM: 44R
242230mE 3009637mN
Elevation: 168.21±2.5 m
Accuracy: 3.79 m
Time: 23-09-2026 09:43
Powered by NoteCam


JCI GROUP
Powered by NoteCam
Latitude: 27.170629
Longitude: 78.403353
Elevation: 172.65±1.05 m
Accuracy: 5.492 m
Time: 25-09-2026 09:47
Note: nala

Drone Survey Commenced in Firozabad on 24th September 2026









11. Photographic Documentation of Field Reconnaissance



Figure 2 Primove team with the survey team and SPOC in the Firozabad City

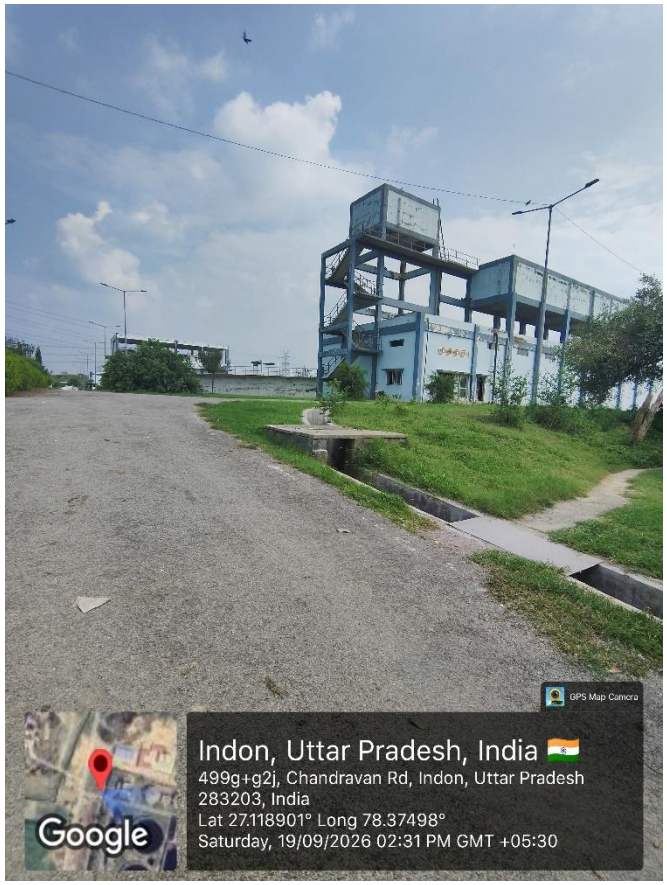
Outfall Locations observed as a part of Reconnaissance Survey





STP's observed as a part of Reconnaissance Survey






GPS Map Camera
Firozabad, Uttar Pradesh, India
5c76+55m, Sallayi Rd, Nagla Mirza Bara, Firozabad,
Uttar Pradesh 283203, India
Lat 27.163388° Long 78.410444°
Thursday, 17/09/2026 11:01 AM GMT +05:30
Google


Latitude: 27.163831
Longitude: 78.405433
Elevation: 68.3m±15m
Time: 17-09-2026 10:45
Note: bypass road Kishan nagar
Powered by NoteCam


Firozabad, Uttar Pradesh, India
5c76+55m, Sallayi Rd, Nagla Mirza Bara, Firozabad,
Uttar Pradesh 283203, India
Lat 27.163362° Long 78.410507°
Thursday, 17/09/2026 10:57 AM GMT +05:30
Google


Firozabad, Uttar Pradesh, India
2, Saanti Road, Sailae's Puliya, Abbas Nagar, Firozabad,
Uttar Pradesh 283203, India
Lat 27.160142° Long 78.414506°
Thursday, 17/09/2026 11:06 AM GMT +05:30
Google







Figure 3 Survey challenges at Satya Nagar, Firozabad: Dense silt and algal bloom obstructing invert level access and Severe blockage at Azad Nagar, Firozabad: Solid waste accumulation preventing survey access



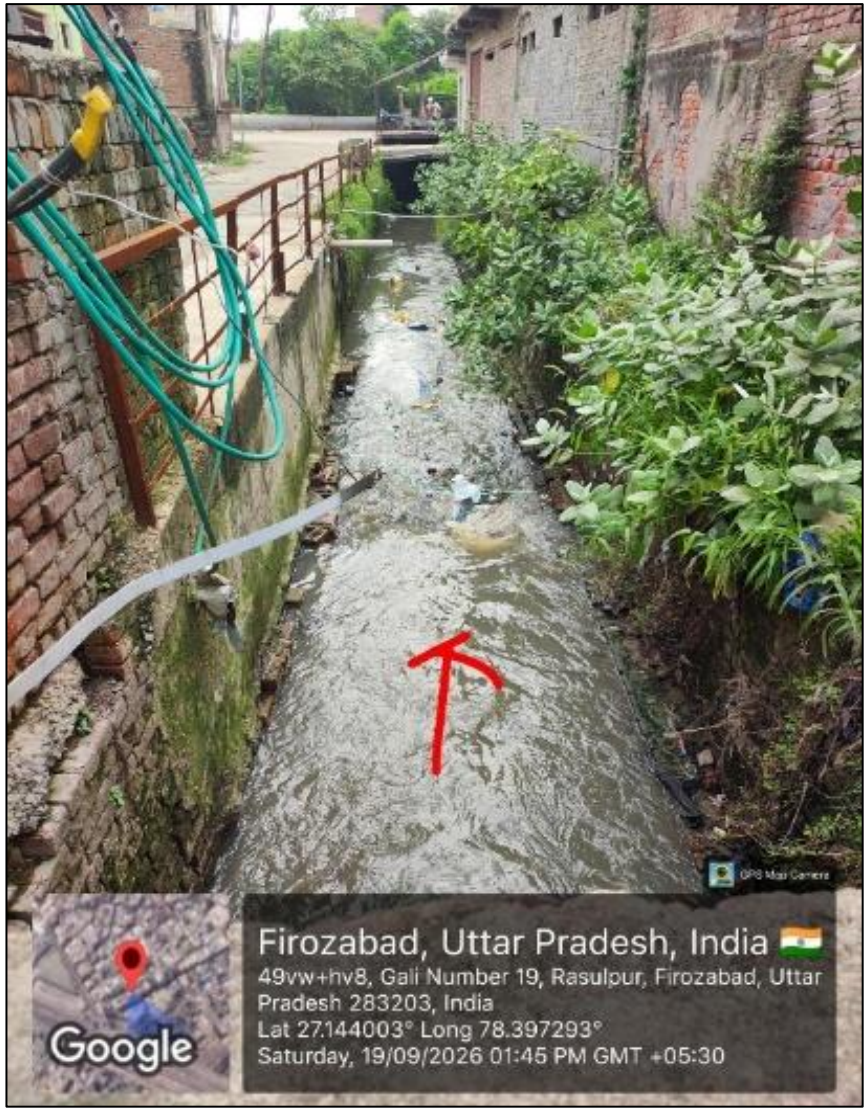
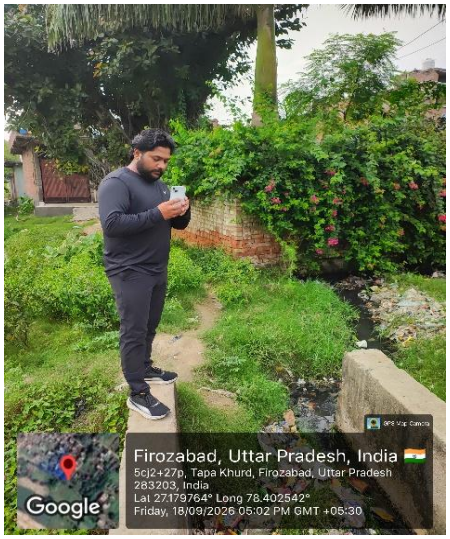


Figure 4 Survey conditions at Sanjay Singh Colony, Firozabad: Vegetation and debris complicating invert access and primary drain at Rasulpur, Firozabad


Firozabad, Uttar Pradesh, India
5cj2+27p, Tapa Khurd, Firozabad, Uttar Pradesh 283203, India
Lat 27.179764° Long 78.402542°
Friday, 18/09/2026 05:02 PM GMT +05:30
Google


Firozabad, Uttar Pradesh, India
5cj2+27p, Tapa Khurd, Firozabad, Uttar Pradesh 283203, India
Lat 27.179768° Long 78.402528°
Friday, 18/09/2026 05:02 PM GMT +05:30
Google


UTM: 44R
240013mE 3005933mN
Elevation: 166.94±2.5 m
Accuracy: 3.79 m
Time: 19-09-2026 16:15
Powered by NoteCam


Firozabad, Uttar Pradesh, India
5cc5+mjm, Nala Rd, Bhikan Pur, Firozabad, Uttar Pradesh 283203, India
Lat 27.1716° Long 78.409086°
Friday, 18/09/2026 04:41 PM GMT +05:30
Google


Firozabad, Uttar Pradesh, India
49mw+3v9 Basat Nale Ki Puliya, Fatehabad Rd, Sanjay Singh Colony, Om Nagar Colony, Firozabad, Uttar Pradesh 283203, India
Lat 27.1324° Long 78.397338°
Saturday, 19/09/2026 02:09 PM GMT +05:30
Google


Firozabad, Uttar Pradesh, India
4cq8+4wf, Nalkoop Colony, Firozabad, Uttar Pradesh 283203, India
Lat 27.137804° Long 78.417194°
Thursday, 17/09/2026 06:10 PM GMT +05:30
Google

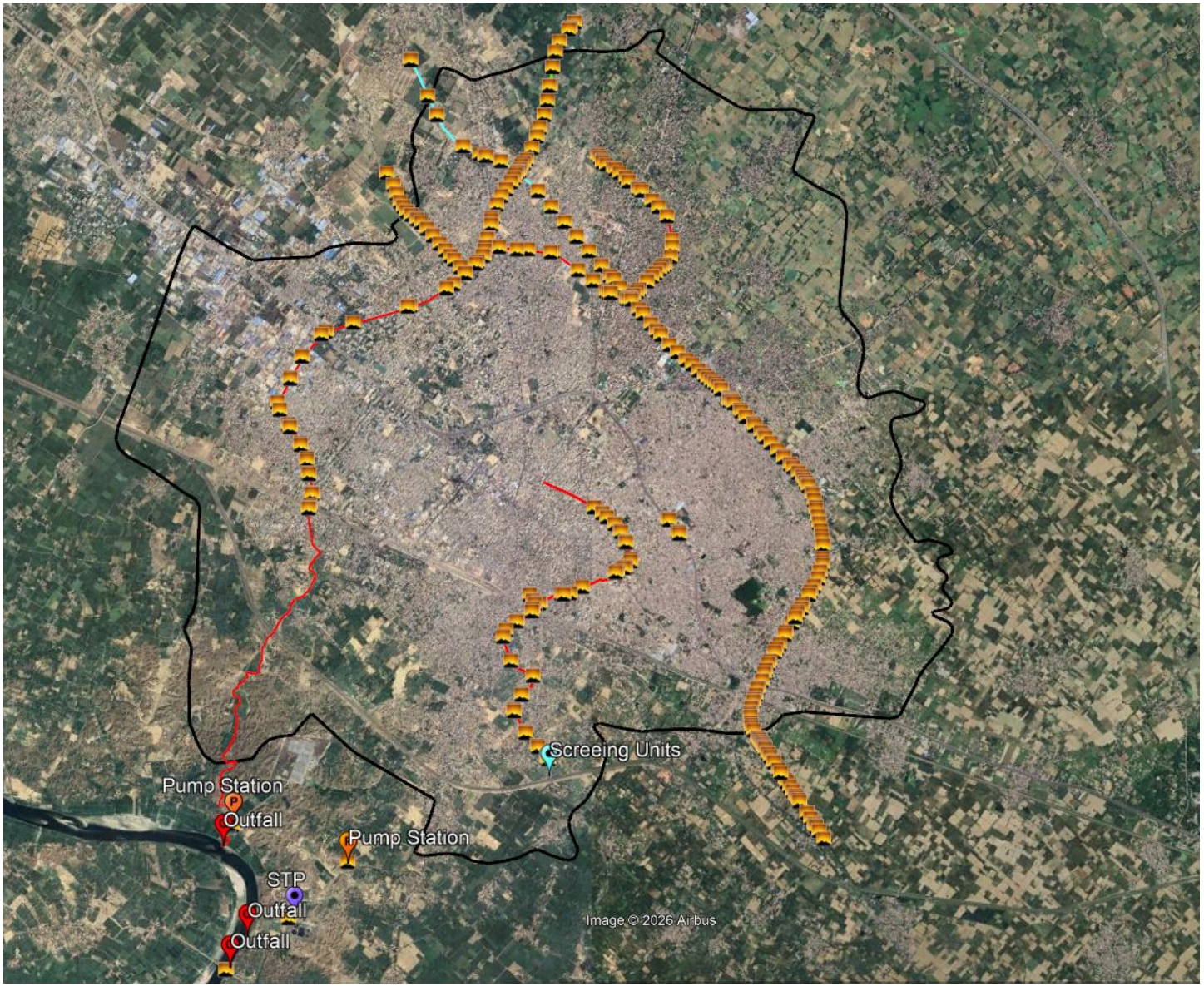


Figure 5 Extent of reconnaissance survey coverage across Firozabad Nagar Nigam